Nishant Chandna

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Education

Delhi Technological University August 2023 – May 2027
B.Tech in Mechanical Engineering with Specialization in Automotive Engineering *9.42 CGPA*

Honors and Awards

- **2nd Position (Self-Funded Category)** in the **DARPA (Defense Advanced Research Projects Agency) Triage Challenge Phase-2**, winning a prize of **$150,000**.
- **2nd Position (Self-Funded Category)** in the **DARPA (Defense Advanced Research Projects Agency) Triage Challenge Phase-1**, winning a prize of **$60,000**.
- **2nd Position** in the simulation stage and **3rd place** in the physical stage of **ICUAS (International Conference on Unmanned Aircraft Systems) 2024**.
- **Research Excellence Award** by Delhi Technological University.

Experience

Founding Researcher May 2026 – Present
FPV Labs, Bangalore *Egocentric SLAM · sim-to-real manipulation*

- Built out the egocentric pose stack: validated commodity ARKit tracking against a 30-camera Vicon rig (8 motion regimes, ∼340 m, centimetre-level ATE at true metric scale), built the Python ATE/RPE, drift and revisit benchmarking suite used to compare classical (ORB-SLAM3, VINS-Fusion, OKVIS2) against feed-forward (DROID-SLAM, MASt3R-SLAM, VGGT-SLAM) pipelines, and shipped a Kalibr-calibrated stereo-inertial estimator at **1.03 cm** median pose error over 3,260 poses.
- Replicated the SimFoundry pipeline end to end to stand up an internal sim-to-real manipulation loop: ∼30 s of phone video into a physics-ready Isaac/OmniGibson replica of the workspace (Gaussian-splat background, estimated mass, friction and collision geometry), then MimicGen-style amplification of ∼10 teleoperated SO-101 demonstrations into thousands of success-verified synthetic episodes.

Voluntary Researcher April 2026 – Present
SAIR Lab, University at Buffalo *Vision-language navigation*

- Integrating the lab's vision-language navigation stack with **OmniNavBench** (RSS 2026; 6 navigation primitives, 3 embodiments, 170 scenes), writing the policy-server bridge that supplies ground-truth odometry and depth-projected scans directly in place of the platform's intrinsics- and TF-dependent depth interface.

Research Intern Feb 2026 – May 2026
RRC, IIIT Hyderabad (Prof. Sourav Garg) *Learned planning costs for navigation*

- Developed **Plann3r**, replacing MASt3R-Nav's pixel-graph plus Dijkstra planner with a learned goal-conditioned costmap predictor: a frozen VGGT-1B backbone supplies per-pixel 3D world points and confidence, encoded into an 8-channel explicit-geometry map (absolute XYZ, goal direction, log-distance, confidence) that a U-Net head regresses into geodesic planning cost.

Research Intern May 2025 – July 2025
RRC, IIIT Hyderabad (Prof. Madhava Krishna) *LiDAR odometry and mapping*

- Implemented degeneracy awareness and localizability analysis to improve robustness in LiDAR odometry and mapping.
- Integrated Scan Context as a backend for place recognition in state-of-the-art ICP algorithms, adding loop closures and factor graph optimization for global consistency in SLAM pipelines, reducing global drift by 40%.

Vice Captain & Autonomy Lead August 2023 – Present
UAS-DTU, Delhi Technological University *Multi-robot autonomy*

- Led the Autonomy Team for the DARPA Triage Challenge, designing and implementing autonomy frameworks that enable robots to navigate, explore, and make decisions in unknown and dynamic environments.
- Developed and integrated perception, planning, and control modules for coordinated multi-robot behavior across both ground and aerial platforms.

Publications

Plann3r: Predicting Planning Costs Grounded in 3D 2026

CoRL 2026 (Under Review)

- 3D grounded method to predict pixel-level planning cost for any given set of images with an arbitrary subgoal pixel.
- Integrated learned costmap prediction with control-conditioned planning, outperforming connectivity-based search planners across diverse navigation tasks.

DAMM-LOAM: Degeneracy-Aware Multi-Metric LiDAR Odometry and Mapping 2025

IROS Active Perception Workshop 2025

- Proposed a novel framework to improve pose estimation in challenging, low-feature environments such as corridors.
- Demonstrated improved performance over state-of-the-art LOAM frameworks in both degenerate and outdoor scenarios.

Autonomous UAV Navigation and Mapping for Accurate Fruit Detection and Counting 2025

International Conference on Unmanned Aircraft Systems (ICUAS)

- Presented a UAV-based fruit counting framework combining YOLOv8 detection, 3D mapping via LiDAR-camera fusion, and TSP-A*-based path planning.
- Achieved 98% accuracy in simulation and 90% in real-world testing; placed 2nd in simulation and 3rd overall at the ICUAS 2024 UAV Competition.

Projects

FairEval SO-101 | *LeRobot, PyTorch, OpenCV, Next.js*

- Built an evaluation harness that makes two real-robot policy checkpoints actually comparable: every checkpoint faces the same counterbalanced object layouts in the same order, restaged against the reference frames the first run captured, removing the layout and framing confounds that otherwise get attributed to the checkpoint.
- Measured what a task-finetuned VLA policy tolerates by sweeping camera intrinsics (focal length up to 2×) alongside mechanism-scoped defocus, lighting and framing conditions over a 161-episode, 129k-frame dataset, with the baseline re-encoded through the identical path so codec generation stays constant.

Drone Based Forest Mapping | *C++, Python, Bash, PX4, ROS, Git*

- Deployed a PX4-based UAV in a simulated forest environment in Gazebo; implemented forest mapping using GenZ-ICP and benchmarked it against KISS-ICP; fused ICP outputs with GPS and IMU data for robust state estimation using dynamic confidence-based weighting.
- Designed and simulated custom environments to systematically evaluate and validate the mapping and localization pipeline.

Collision Avoidance using EGO-Planner and VINS-Mono | *C++, VIO, PX4, ROS*

- Implemented GPS-denied autonomous navigation and collision avoidance in indoor environments using VINS-Mono for visual-inertial state estimation and EGO-Planner for real-time local trajectory generation.
- Integrated perception, planning, and control modules on a lightweight quadrotor running PX4 and ROS, enabling agile flight in cluttered 3D spaces with high-speed obstacle avoidance.

Reward Function Improvement in PyFlyt (RL for UAVs) | *RL, Stable Baselines 3, Gymnasium*

- Contributed to a reinforcement learning environment repository by enhancing the reward function, introducing a negative reward based on the rate of change of orientation over previous time steps to improve drone stability in scenarios such as hovering and pole balancing, significantly improving efficiency and convergence speed.

Technical Skills

Robotics & Simulation: ROS/ROS 2, PX4, Gazebo, Isaac Sim, Isaac Lab, OmniGibson, ArduPilot SITL, LeRobot, Docker, Git
Perception & Estimation: SLAM, VIO/LIO, Sensor Fusion, Factor Graph Optimization (GTSAM), Kalibr Calibration, Gaussian Splatting, 3D Reconstruction
Learning: Python, PyTorch, Imitation Learning, Vision-Language Navigation, Vision-Language-Action Policies, Reinforcement Learning (Stable Baselines 3)
Hardware: Jetson Nano, Jetson Orin, Pixhawk, Arduino, LiDARs, Intel RealSense, OAK-D, SO-101 Arms